Week 6 question sheet

Problem sheet

Instructions

Attempt all 11 questions and all 16 scorable parts. The target time is 270 minutes and the total is 100 marks. Show column gradients, Hessians, distributional assumptions, standardisations and every condition needed for a local, global, exact or approximate conclusion. A basic scientific calculator is permitted only for the final decimal in `W06-Q06-PB`; the exact answer is sufficient. No software is needed.

Use hints.md only after a serious attempt. Do not consult solutions.md until you have completed or paused every part.

Question 1 — necessary and sufficient conditions

Let

$$f(x,y) = x^2 + xy + y^2 - 6x, \qquad (x,y) \in \mathbb{R}^2.$$

`W06-Q01-PA` **(6 marks; 16 minutes)** Find every stationary point. Compute the Hessian and use it to classify each stationary point locally.

`W06-Q01-PB` **(4 marks; 11 minutes)** Explain why solving $\nabla f(\mathbf{x}) = \mathbf{0}$ alone would give only a candidate. Then justify whether your local classification from Part A is also a unique global classification.

Question 2 — a strictly concave objective

Let

$$g(x,y) = 8x + 6y - x^2 - y^2 - xy, \qquad (x,y) \in \mathbb{R}^2.$$

`W06-Q02-PA` **(7 marks; 21 minutes)** Find every stationary point and classify it using the Hessian.

`W06-Q02-PB` **(5 marks; 13 minutes)** State the conditions that allow a global conclusion on the stated domain. Prove that the point found in Part A is the unique global maximiser and calculate the maximum value.

Question 3 — a local maximum and a saddle point

`W06-Q03-PA` **(6 marks; 20 minutes)** For

$$h(x,y) = x^3 - 3x - y^2, \qquad (x,y) \in \mathbb{R}^2,$$

find and classify every stationary point. Decide whether h has a global maximum or a global minimum on $\mathbb{R}^2$, giving a reason rather than relying only on the Hessian at the candidates.

Question 4 — when the Hessian test is inconclusive

`W06-Q04-PA` **(8 marks; 22 minutes)** Let

$$p(x,y) = x^4 + y^4, \qquad (x,y) \in \mathbb{R}^2.$$

Find the stationary points and evaluate the Hessian there. Explain why the usual definite-Hessian test is inconclusive. Then classify the point by a direct argument, including whether the conclusion is local or global and whether the optimiser is unique. State whether a global maximum exists.

Question 5 — parameter-dependent curvature

`W06-Q05-PA` **(10 marks; 25 minutes)** For $a \in \mathbb{R}$, define

$$q_a(x, y) = -x^2 - 2axy - y^2, \qquad (x, y) \in \mathbb{R}^2.$$

Classify the curvature of q_a for $|a| < 1$, $|a| = 1$ and $|a| > 1$. In each case, classify the origin and describe the complete set of global maximisers, or explain why no global maximum exists. Your answer must distinguish negative definiteness, negative semidefiniteness and indefiniteness.

Question 6 — sample mean and sample variance

`W06-Q06-PA` **(5 marks; 13 minutes)** The observations $2, 4, 4, 6$ are realised values of a random sample $X_1, \dots, X_4$. Calculate the observed sample mean $\bar{x}$ and the observed sample variance

$$s^2 = \frac{1}{4-1} \sum_{i=1}^{4} (x_i - \bar{x})^2.$$

Distinguish these observed values from the random variables $\bar{X}$ and S^2.

`W06-Q06-PB` **(8 marks; 19 minutes)** Suppose instead that

$$X_1, \dots, X_{10} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 4),$$

where the second normal parameter is the variance. State the sampling distribution of $9S^2/4$. Given that

$$\mathbb{P}(Q \leq 16.919) = 0.95 \quad \text{for } Q \sim \chi^2_9,$$

find c such that $\mathbb{P}(S^2 \leq c) = 0.95$. Give an exact expression; a basic calculator may be used to report a two-decimal approximation. State the assumption needed for this exact chi-squared result.

Question 7 — exact and approximate distributions of a sample mean

`W06-Q07-PA` **(7 marks; 17 minutes)** Suppose

$$X_1, \dots, X_{16} \overset{\text{iid}}{\sim} \mathcal{N}(20, 16).$$

Find the exact distributions of $\bar{X}$ and $T := \sum_{i=1}^{16} X_i$. Using $\Phi(1) = 0.8413$ and $\Phi(2) = 0.9772$, calculate

$$\mathbb{P}(18 < \bar{X} < 22) \quad \text{and} \quad \mathbb{P}(T > 336),$$

where Φ is the CDF of $\mathcal{N}(0, 1)$.

`W06-Q07-PB` **(7 marks; 19 minutes)** Let

$$Y_1, \dots, Y_{100} \overset{\text{iid}}{\sim} \text{Bernoulli}(0.5),$$

with support $\{0,1\}$. Use the central limit theorem, without a continuity correction, to approximate

$$\mathbb{P}(0.4 \le \bar{Y} \le 0.6).$$

Use $\Phi(2) = 0.9772$. State the mean and variance used, show the correct standardisation, and explain why the calculation is approximate rather than exact.

Question 8 — concentration and limiting shape

`W06-Q08-PA` **(6 marks; 21 minutes)** Let $X_1, \dots, X_n$ be i.i.d. with $\mathbb{E}[X_i] = 50$ and $\mathrm{Var}(X_i) = 25$. Use

$$\mathbb{P}(|\bar{X} - \mu| \ge \varepsilon) \le \frac{\sigma^2}{n\varepsilon^2}$$

to find the smallest integer n for which this bound is at most 0.04 when $\varepsilon = 1$. Explain how the bound supports the law of large numbers without asserting a normal sampling distribution.

`W06-Q08-PB` **(5 marks; 13 minutes)** A student writes: "The law of large numbers and the central limit theorem both say that $\bar{X}$ becomes normal and equals μ when n is large." Correct the statement. Your answer must identify the type of convergence supplied by the law of large numbers, the correctly standardised quantity in the central limit theorem, and what neither result says about the individual observations.

Cumulative retrieval

Complete Questions 9–11 without consulting earlier notes.

Question 9 — column gradient and implicit-differentiation retrieval

`W06-Q09-PA` **(5 marks; 12 minutes)** Let $F : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$F(x, \theta) = x^2 + \theta x - 3.$$

Verify that $F(1, 2) = 0$. Compute the column gradient of F at $(x, \theta) = (1, 2)$ and state its dimension. Check the condition that locally permits the equation $F(x, \theta) = 0$ to define $x = x(\theta)$. Then use the total differential to calculate $\mathrm{d}x/\mathrm{d}\theta$ at $(1, 2)$.

Question 10 — definiteness retrieval

`W06-Q10-PA` **(5 marks; 14 minutes)** Classify the quadratic form

$$k(u, v) = 2u^2 + 2uv + 3v^2$$

by writing its symmetric matrix and applying a valid definiteness test. State the implication for the curvature of k on $\mathbb{R}^2$.

Question 11 — conditional moments retrieval

`W06-Q11-PA` **(6 marks; 14 minutes)** Let $X \sim \text{Exponential}(2)$, where 2 is the rate and the support is $x \ge 0$. Independently, let $B \sim \text{Bernoulli}(1/2)$, with support $\{0, 1\}$, and define $Y = 3X + B$. Calculate $\mathbb{E}[Y \mid B]$ as a random variable. Use iterated expectation to obtain $\mathbb{E}[Y]$, then use the law of total variance to calculate $\mathrm{Var}(Y)$.

Self-test questions

Set aside 25 minutes and attempt all four parts without the reading guide, worked examples, hints or exercise solutions. No calculator is needed. Show the condition that supports every local, global, exact or approximate conclusion. The self-test has 20 marks and does not contribute to a checkpoint or mock-examination score.

Questions

`W06-ST01-PA` — a global maximum (`M9`; 5 marks; 6 minutes) Let

$$f(x, y) = 4x + 2y - x^2 - 2y^2, \qquad (x, y) \in \mathbb{R}^2.$$

Find the stationary point, classify it locally, and justify whether it is the unique global maximiser. Calculate the maximum value.

`W06-ST02-PA` — an inconclusive Hessian (`M9`; 5 marks; 6 minutes) Let

$$g(x, y) = x^4 - y^4.$$

Classify the origin. Your answer must evaluate the gradient and Hessian there and then use a direct sign argument. State whether a global maximum or minimum exists on $\mathbb{R}^2$.

`W06-ST03-PA` — observed sample summaries (`S6`; 5 marks; 6 minutes) For the realised sample $1, 3, 5, 7$, calculate $\bar{x}$ and the course sample variance s^2. State the denominator convention and distinguish the observed values from $\bar{X}$ and S^2.

`W06-ST04-PA` — an exact sampling distribution (`S6`; 5 marks; 7 minutes) Suppose

$$X_1, \ldots, X_9 \overset{\text{iid}}{\sim} \mathcal{N}(5, 9).$$

Find the exact distribution of $\bar{X}$ and calculate $\mathbb{P}(4 < \bar{X} < 6)$, using $\Phi(1) = 0.8413$.